

Tying the Data Knot

A State-of-the-Art Machine
Learning Pipeline for Digital
Romance (V8 Patched Edition)

OCC6 Group 3

WIA1006 Machine Learning

Faculty of Computer Science and
Information Technology

The Hypothesis & Stakes



The Problem

Superficial
Matchmaking

Confounding
Sociological Factors

High Churn &
Ghosting Rates



The Objective

Transitioning from
Predictive to Prescriptive AI

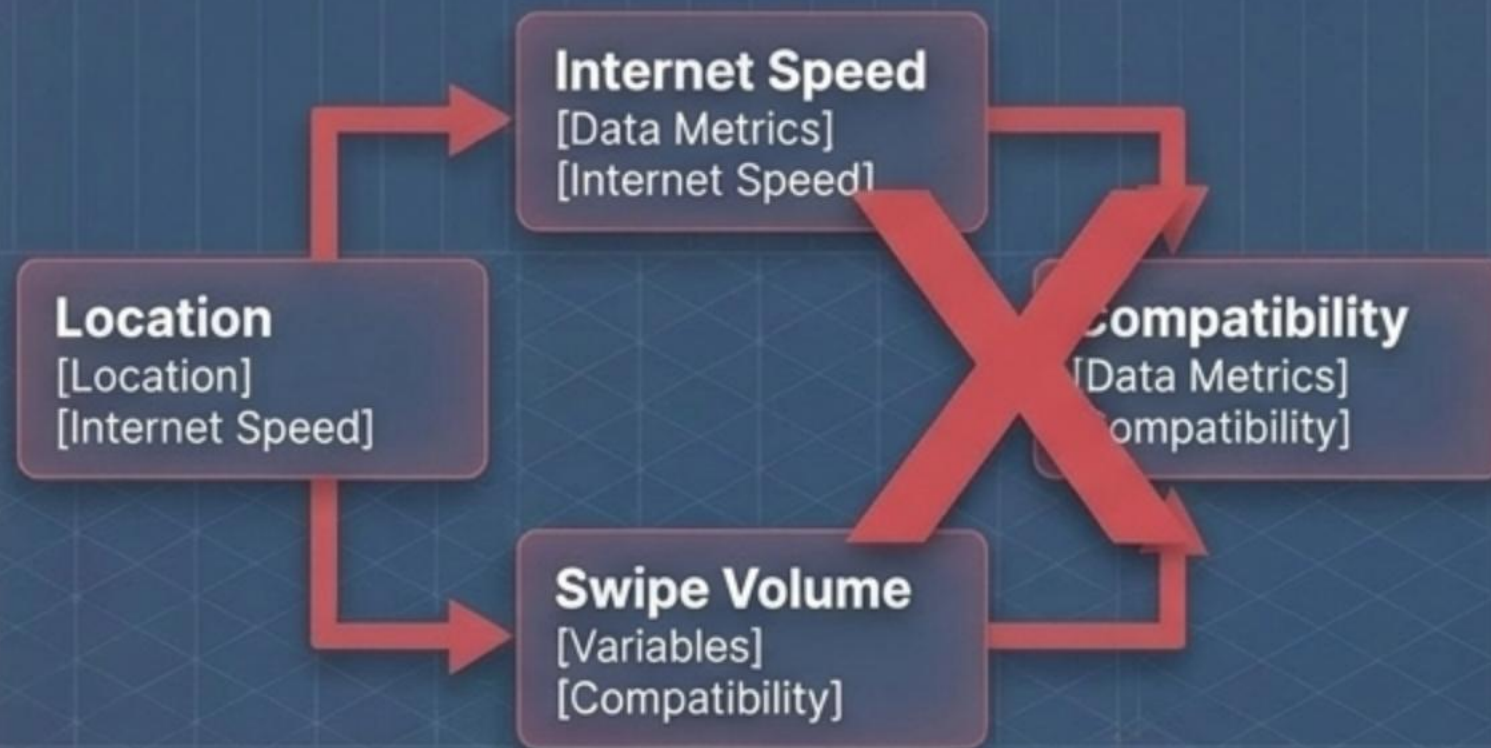
Build a Leakage-Free
Causal Audit Sandbox

Determine Why Matches
Happen, Not Just If

The Illusion of Correlation in Digital Matchmaking

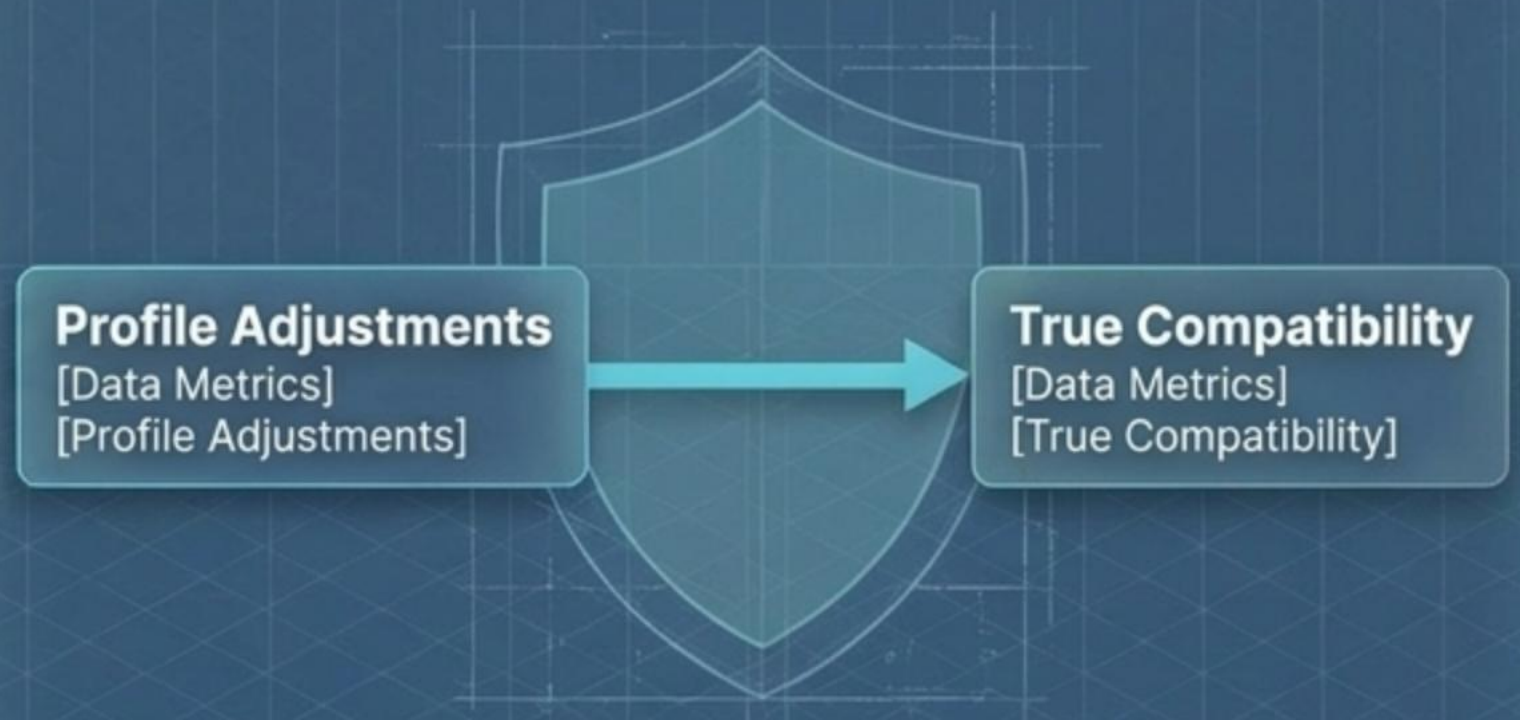
The Problem: High churn and ghosting driven by traditional models conflating spurious correlations (e.g., demographic wealth) with true compatibility.

Predictive ML (The Status Quo)



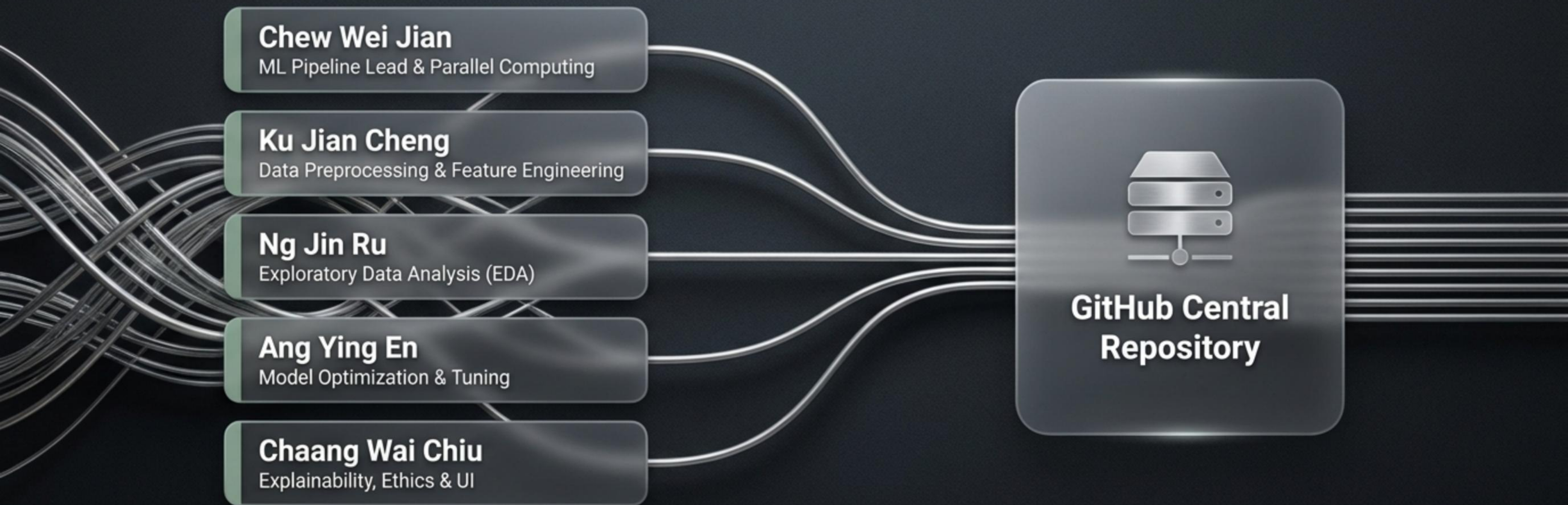
The Confounding Loop: Static demographic features are heavily mediated by latent confounders.

Prescriptive Causal AI (Our Objective)



The Architectural Objective: Transition from prediction to prescription. Build a leakage-free causal audit sandbox to estimate the true interventional probability of a match.

Team Organization & Computing Architecture



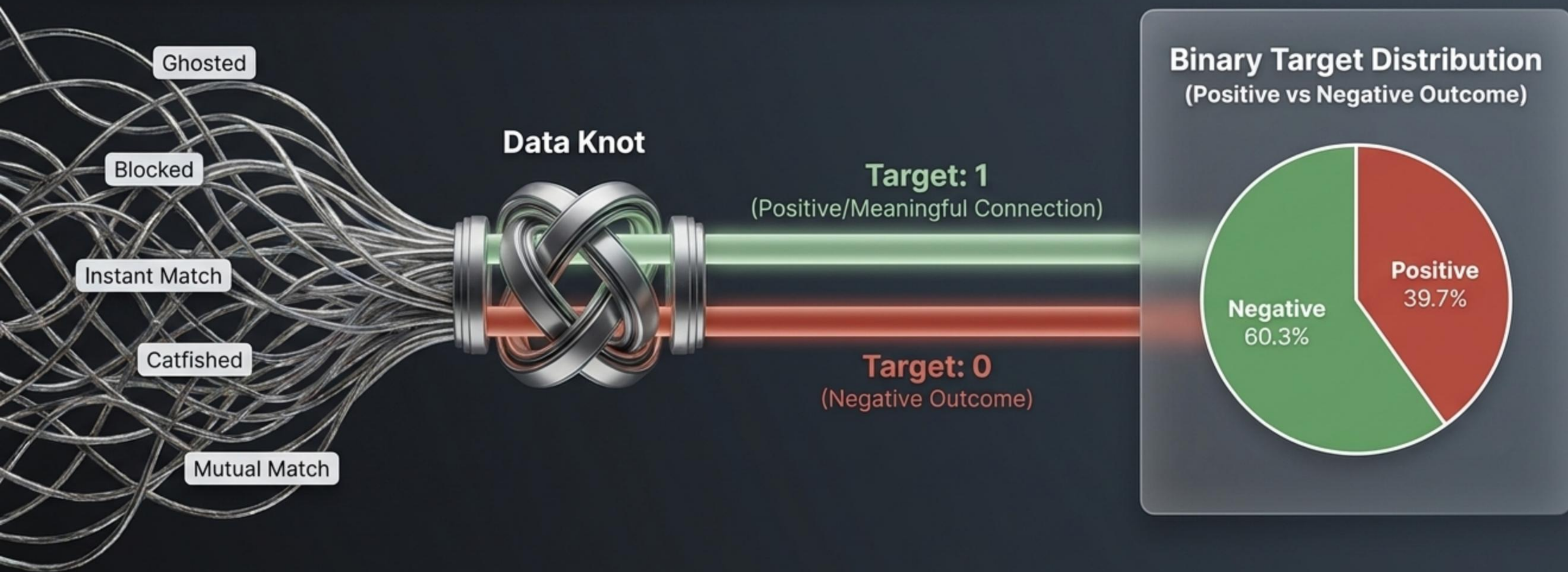
Execution Architecture: Modular Jupyter Notebook Branches | Parallel Execution | Strict Peer-Review Protocol

The Dataset & Target Variable

50,000 Samples

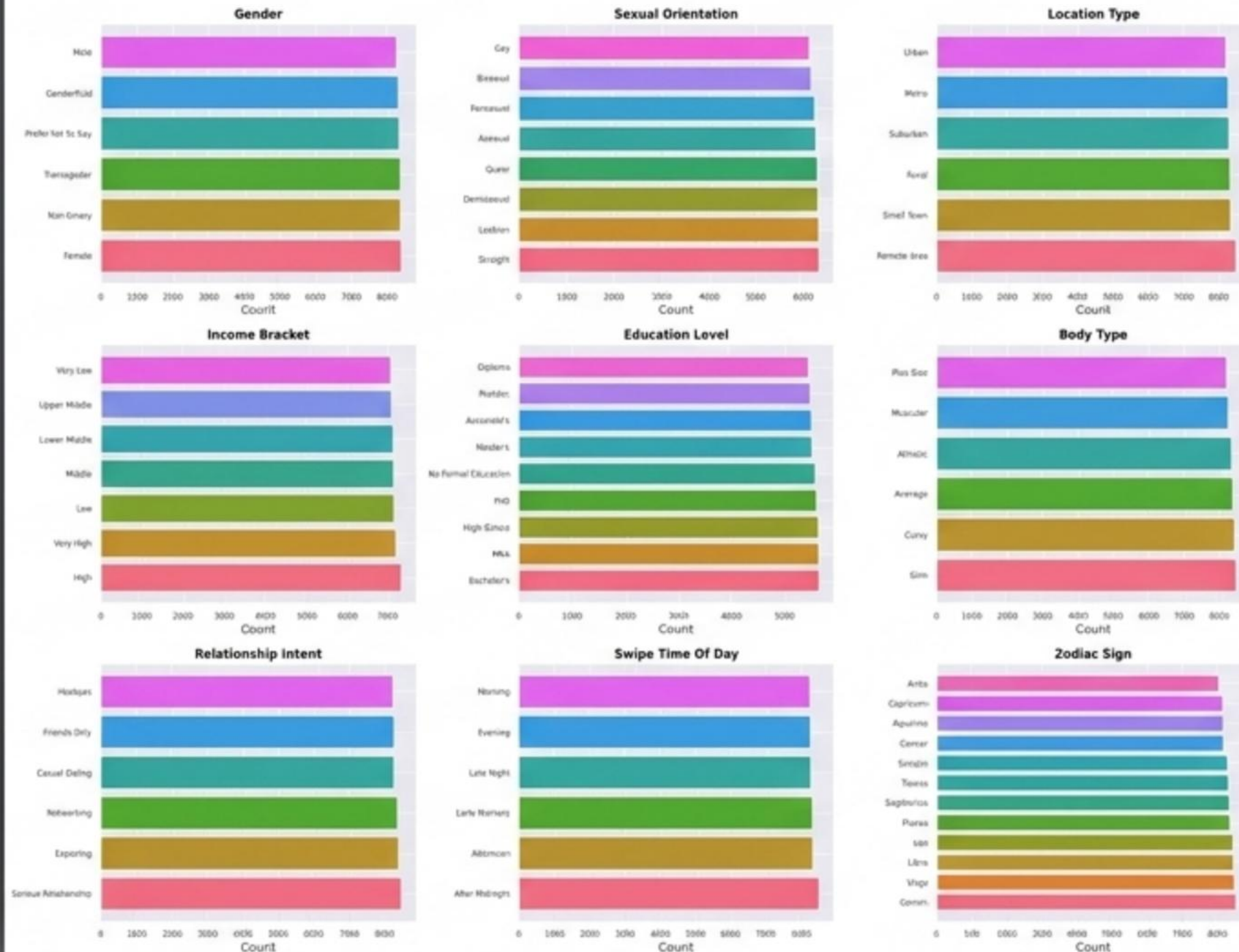
25 Features

Zero Missing Values



Exploratory Data Analysis: Categorical Uniformity

Categorical Feature Distributions



Programmatic Uniformity

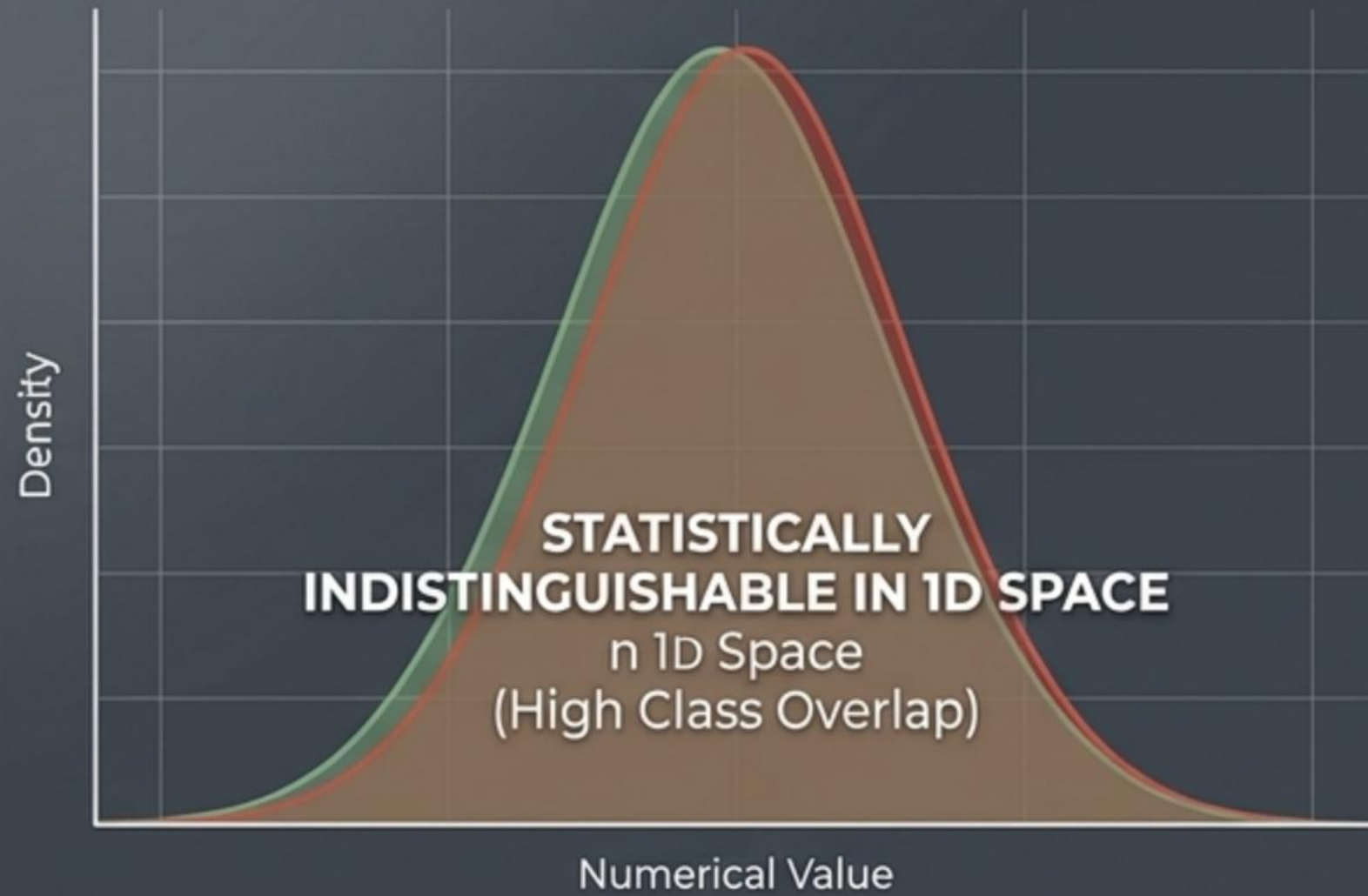
Uniform distributions across sensitive demographic variables.

Prevents class dominance in the algorithm's loss function.

Establishes a pristine, mathematically sound baseline for Demographic Parity Fairness Audits.

Exploratory Data Analysis: Numerical Density & Outliers

Numerical Feature Density Overlap

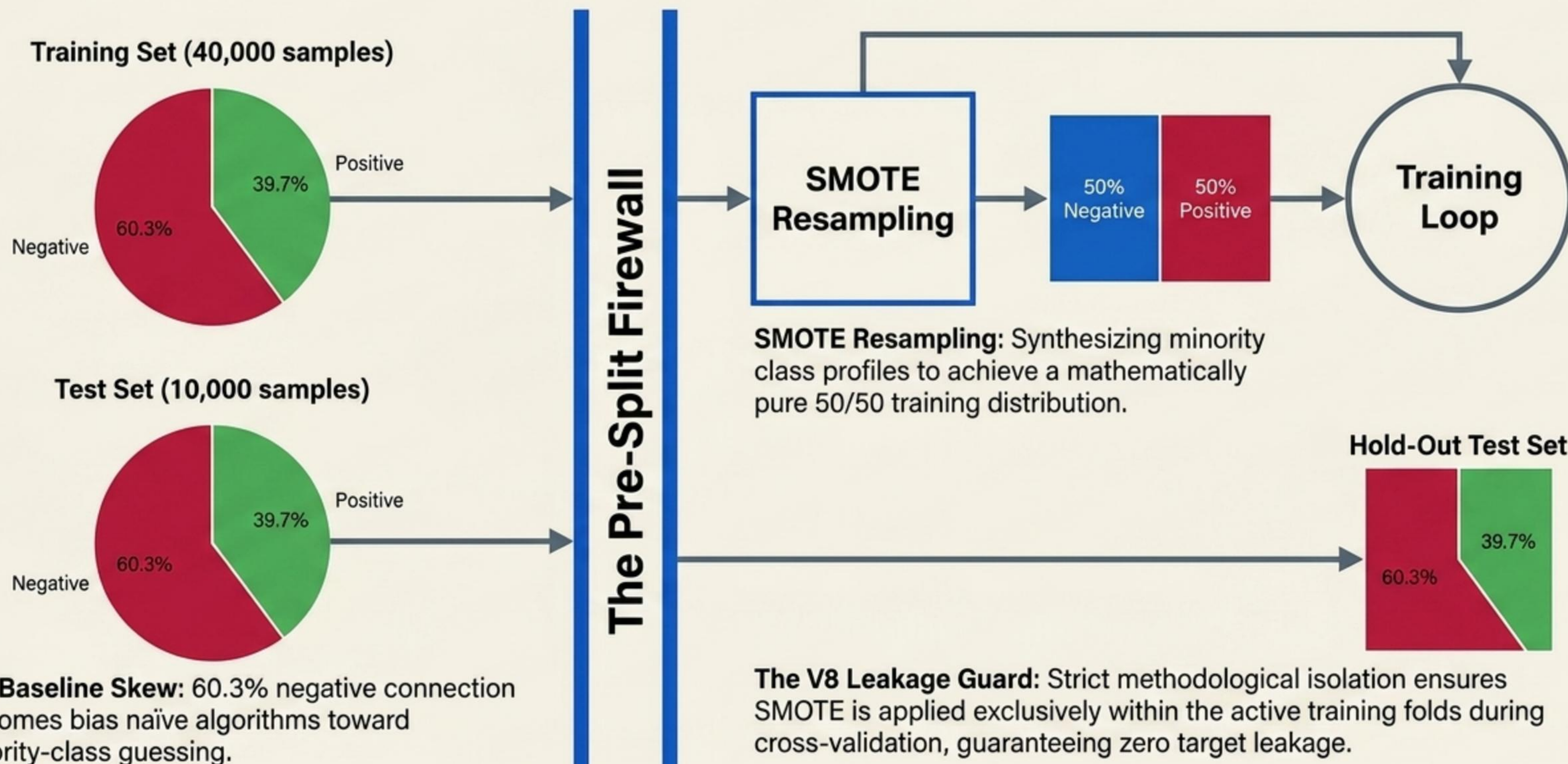


Outlier Detection — Boxplots



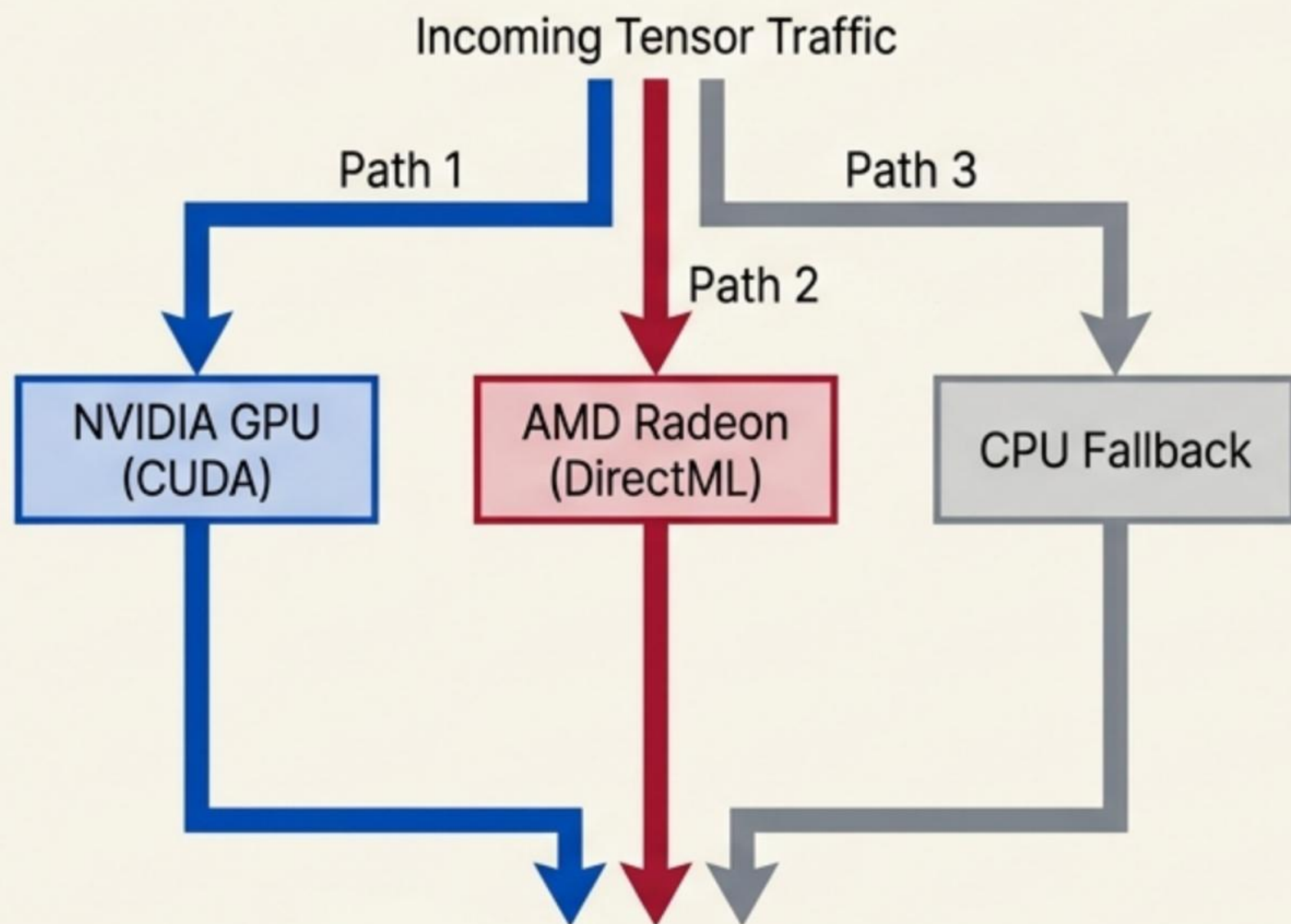
METHODOLOGICAL CONSEQUENCE: UPGRADE TO MEDIAN-BASED ROBUSTSCALER
(Deferred Post-Split to Prevent Leakage)

Protecting the Ground Truth of Matchmaking Data



Engineering a Crash-Proof Engine for Hardware Execution

The Hardware Auto-Detection Engine

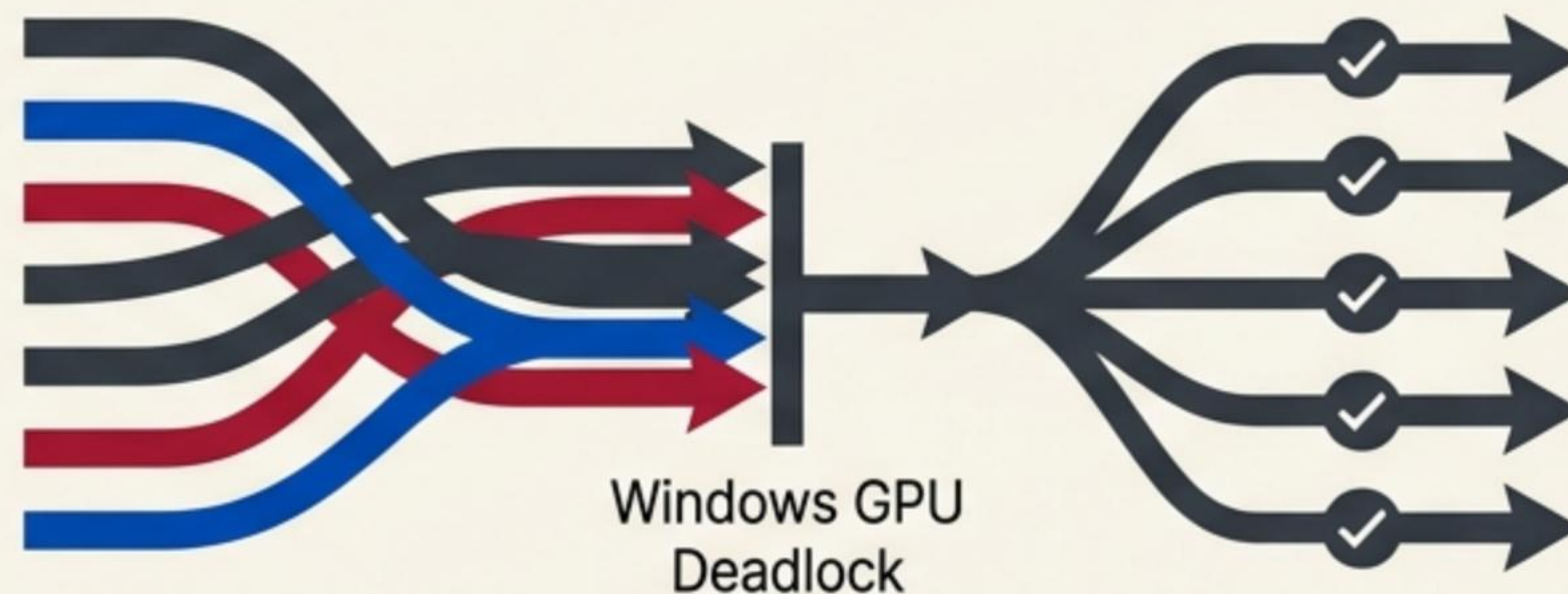


Dynamic Auto-Detection: Real-time tensor routing to NVIDIA CUDA or AMD DirectML ensures 100% self-healing, cross-device execution.

The Concurrency Fix

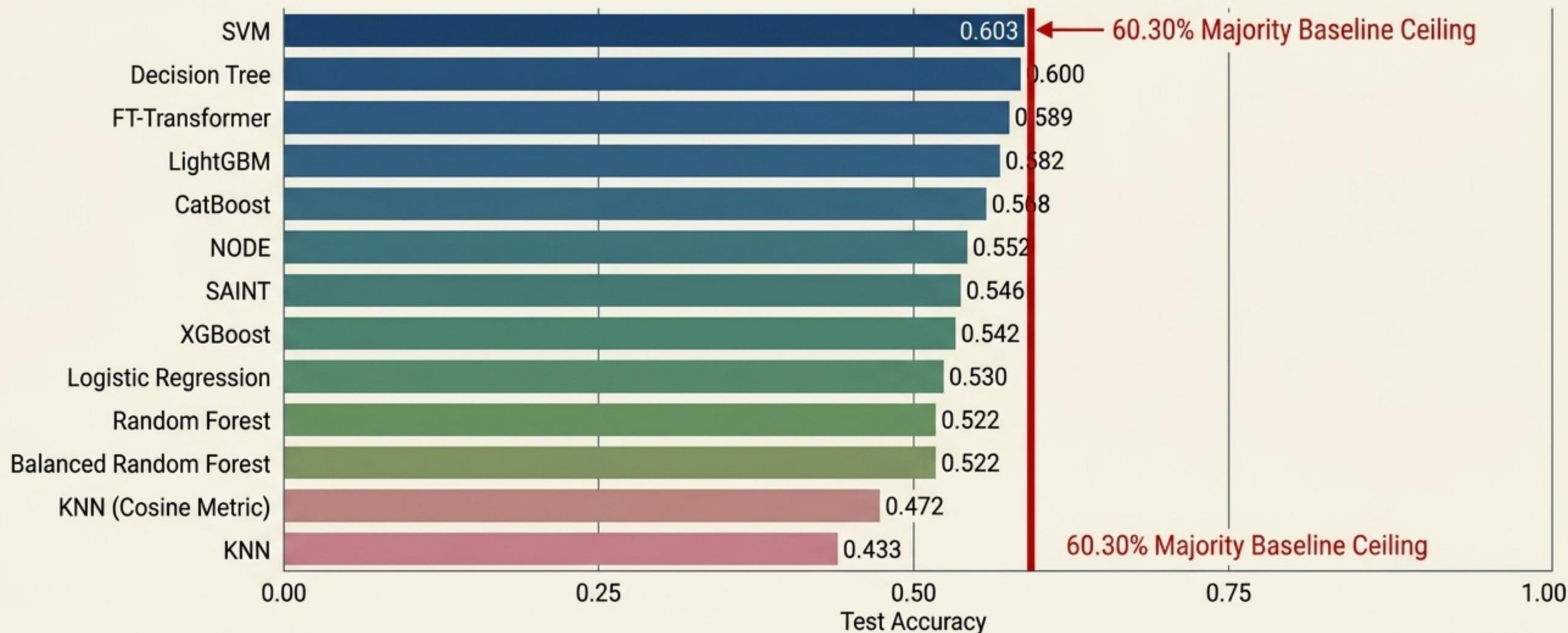
Cross-Validation Loop

WARNING: `n_jobs=-1` → 100% Utilization GPU Crash
FIX APPLIED: `n_jobs=1` (Sequential Outer Isolation)



- **The Deadlock Dilemma:** Concurrent GPU context initializations (under `n_jobs=-1`) cause severe driver lock-ups during cross-validation.
- **Sequential Outer Isolation:** Forcing `n_jobs=1` on outer CV loops guarantees clean, sequential GPU access while unlocking **estimators** to fully parallelize internally.

Pushing 14 Classifiers to the Mathematical Limit



AutoML Benchmarking

- PyCaret and FLAML deployed to establish robust, unbiased algorithmic ceilings.

The Traditional Arsenal

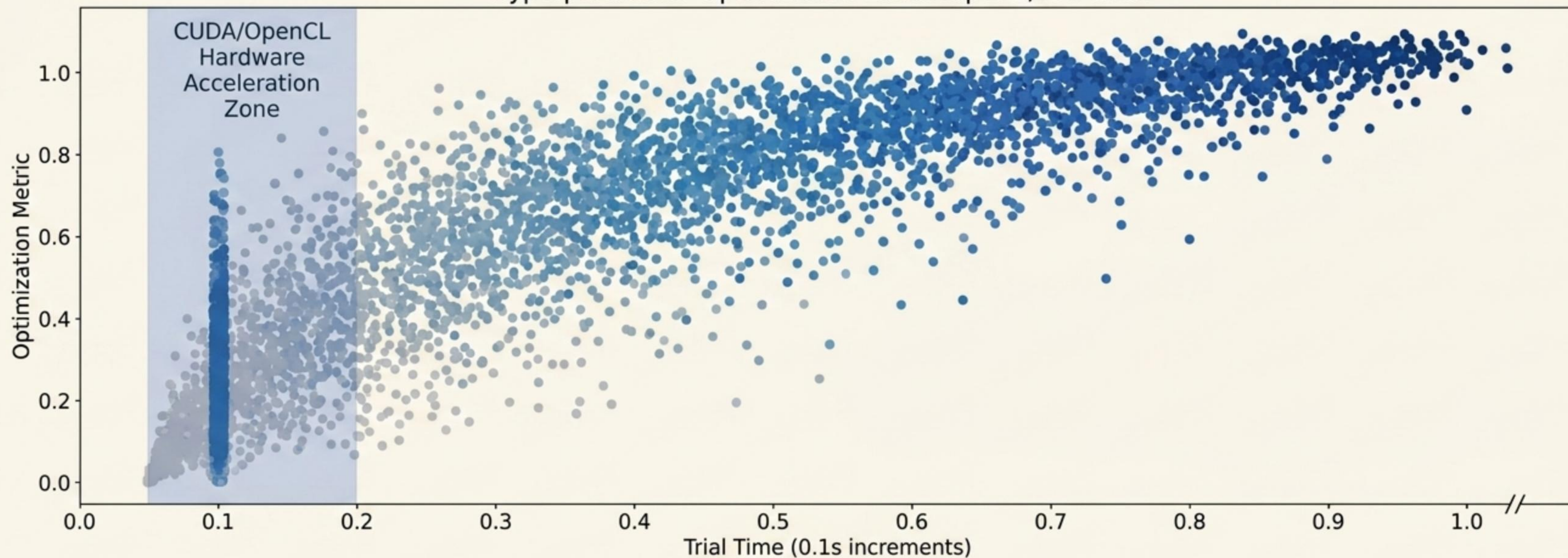
12 additional architectures evaluated, covering linear boundaries, boosting ensembles, and multi-threaded SVM bagging.

The Convergence Phenomenon

Despite extreme mathematical variances in approach, performance hits a rigid ceiling at ~60.30% accuracy—the exact baseline ratio.

Accelerating 1,000 Optimization Trials to Under Four Minutes

Hyperparameter Optimization Landscape: 1,000 Trials



The Grid Search Bottleneck

Traditional hyperparameter tuning is catastrophically slow when exploring deep tree constraints.

Massive Optuna Sweep

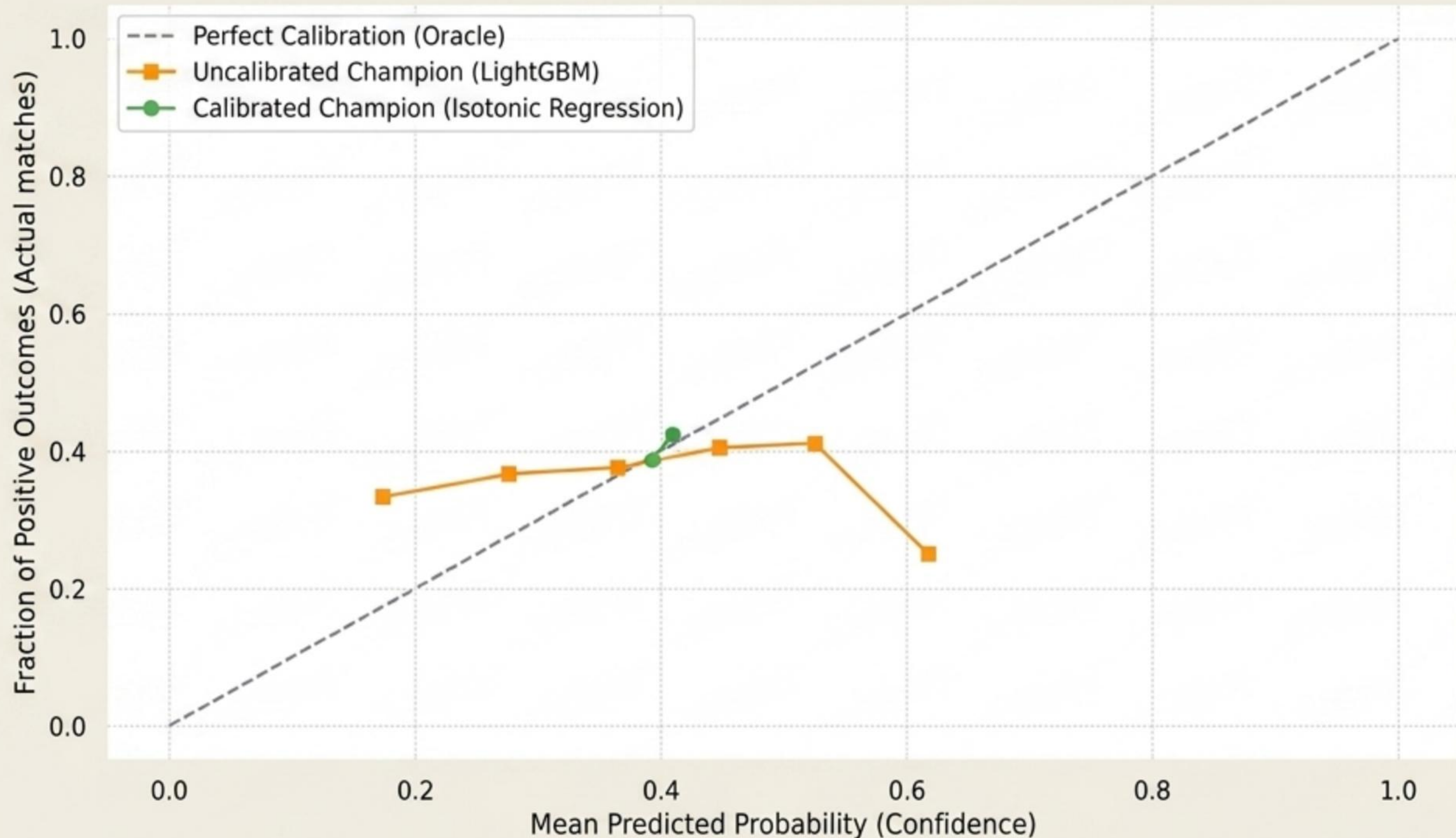
Executing a **1,000**-trial, fully randomized multi-objective optimization matrix.

CUDA/OpenCL Offloading

Routing trial-fitting calculations directly to graphics processing cores compresses individual trial latency down to **0.1 seconds**.

Aligning Raw Model Output with Empirical Reality

□ **Probability Calibration Curve (Reliability Diagram)**



The Calibration Gap

Raw output scores from tree-based algorithms are distorted margins, misrepresenting actual real-world probabilities.

Isotonic Regression

A strict, non-parametric mathematical mapping function applied globally over the Champion Model outputs.

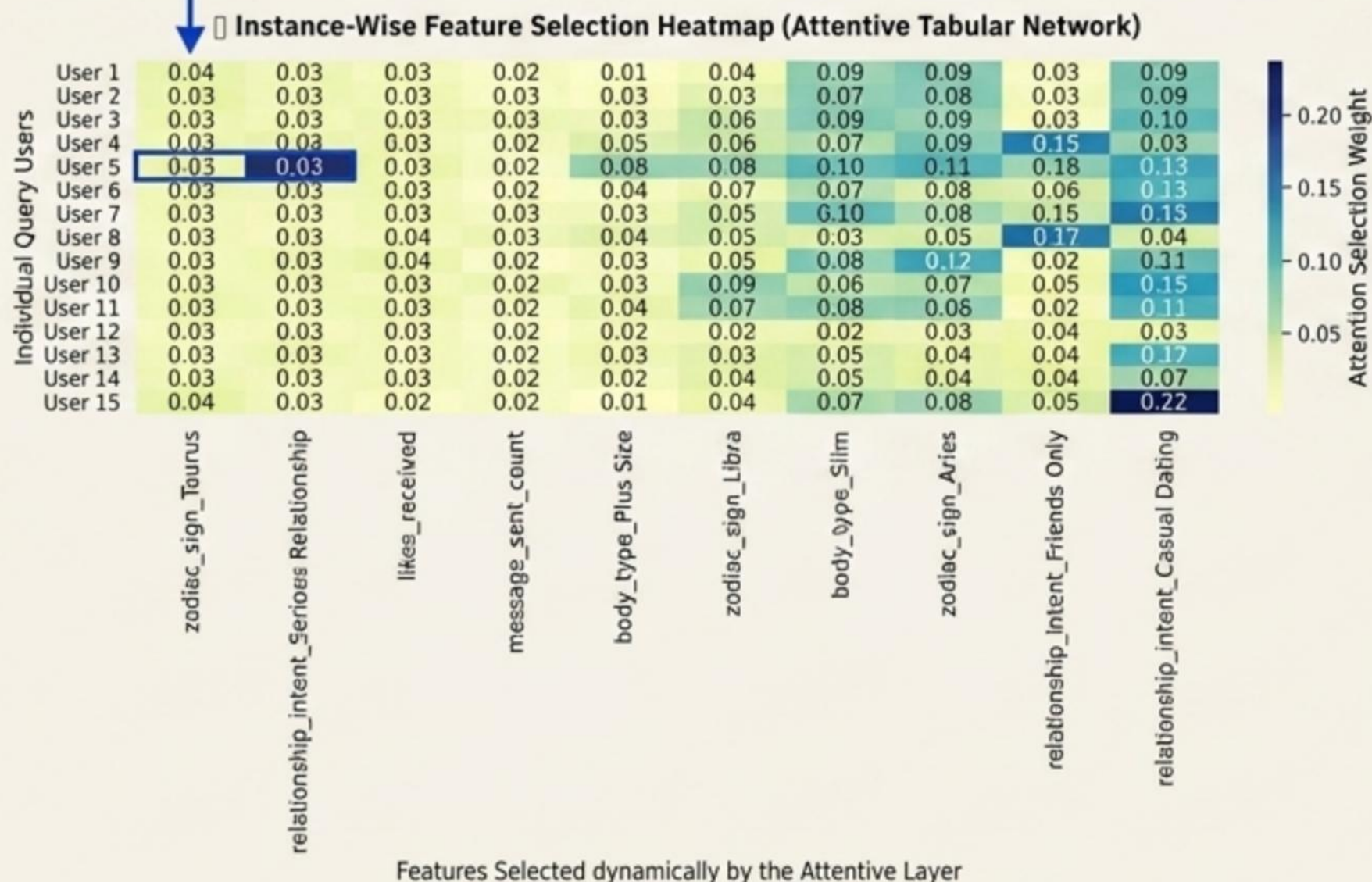
Minimizing the Brier Score

Forcing predicted algorithmic confidence to perfectly mirror the empirical rate of connection success, guaranteeing trustworthy UI thresholds.

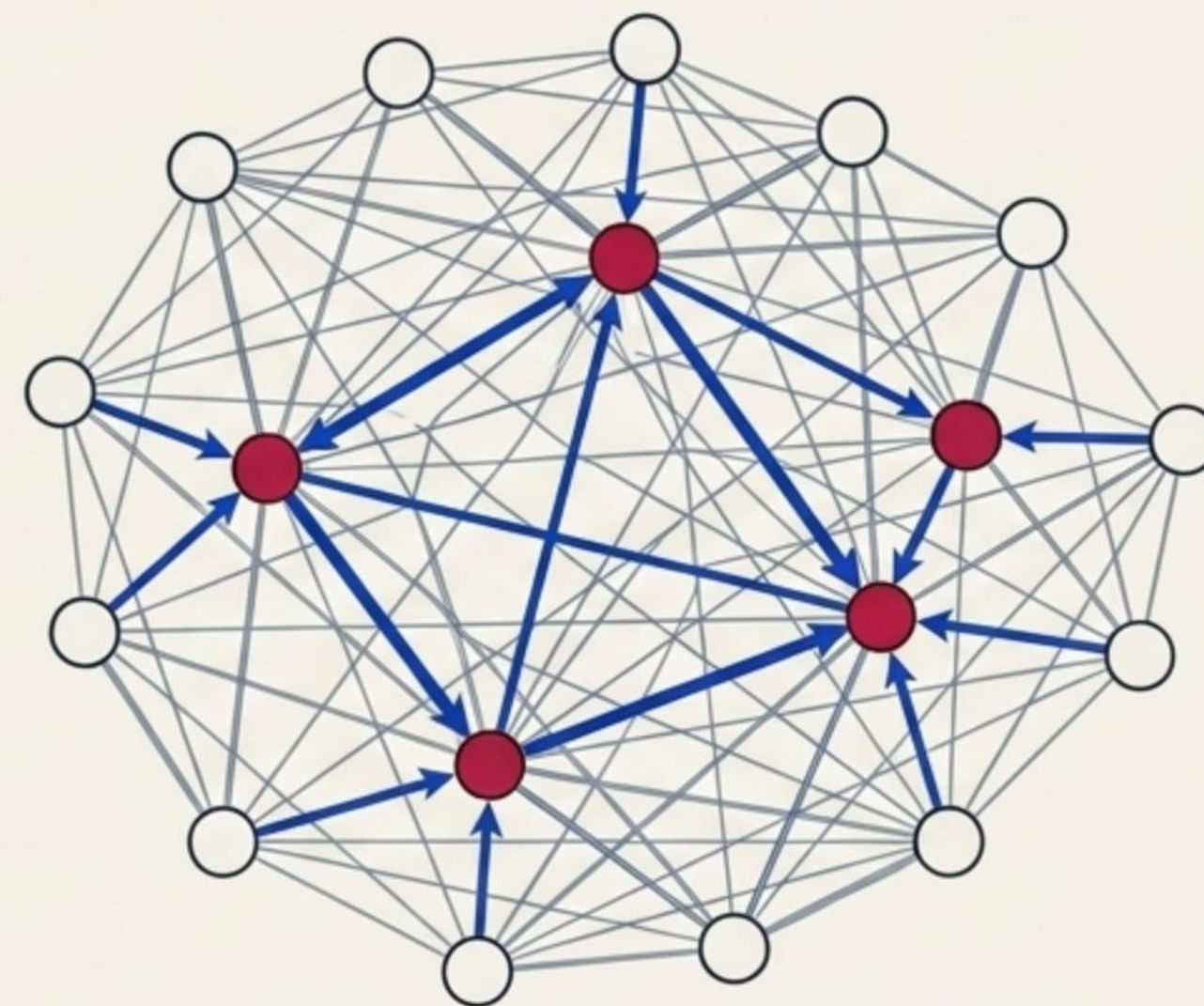
Searching for Non-Linear Synergies in Romance

PyTorch Attentive Tabular Network

Soft-Mask Routing: The network dynamically shifts priority across profile traits differently for every single user query.



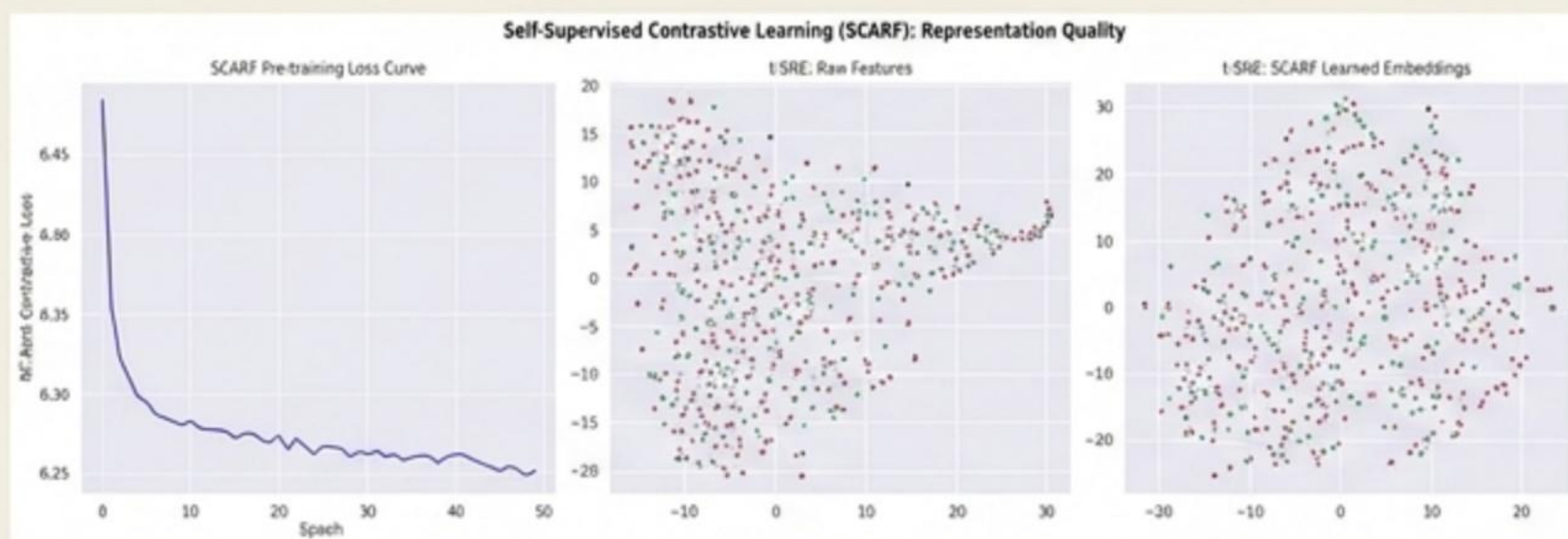
Graph Attention Networks (GAT)



Reframing the tabular problem as **semi-supervised node classification**, mapping **non-linear user-to-user** similarity graphs.

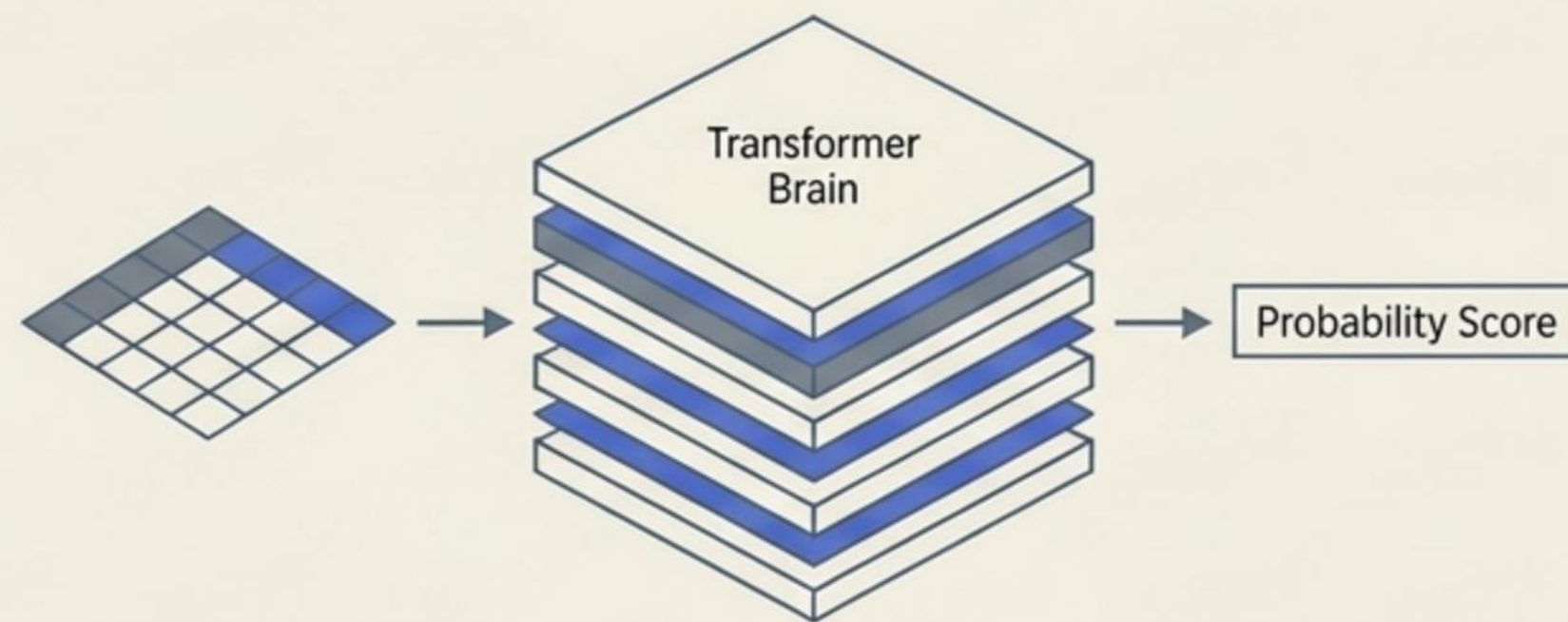
Transductive Pre-Training and Massive Priors

SCARF Contrastive Learning



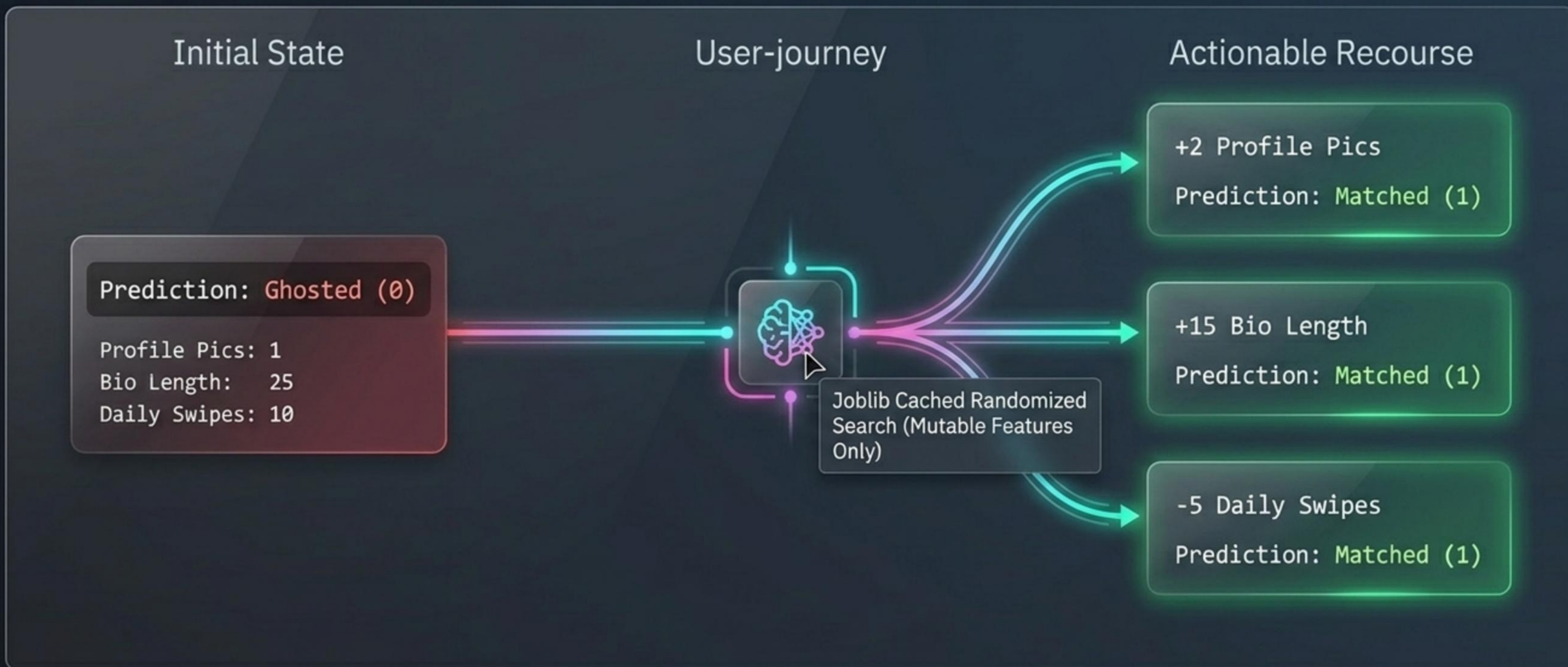
t-SNE Representation: Latent space visualization reveals homogenous, overlapping clusters regardless of successful or failed match outcomes. Self-supervised pre-training via random feature corruption fails to force separation.

TabPFN Zero-Shot Evaluation



TabPFN Zero-Shot Evaluation: Bypassing gradient descent entirely using a Bayesian transformer pre-trained on millions of synthetic datasets. It is the ultimate oracle, yet arrives at the exact same conclusion.

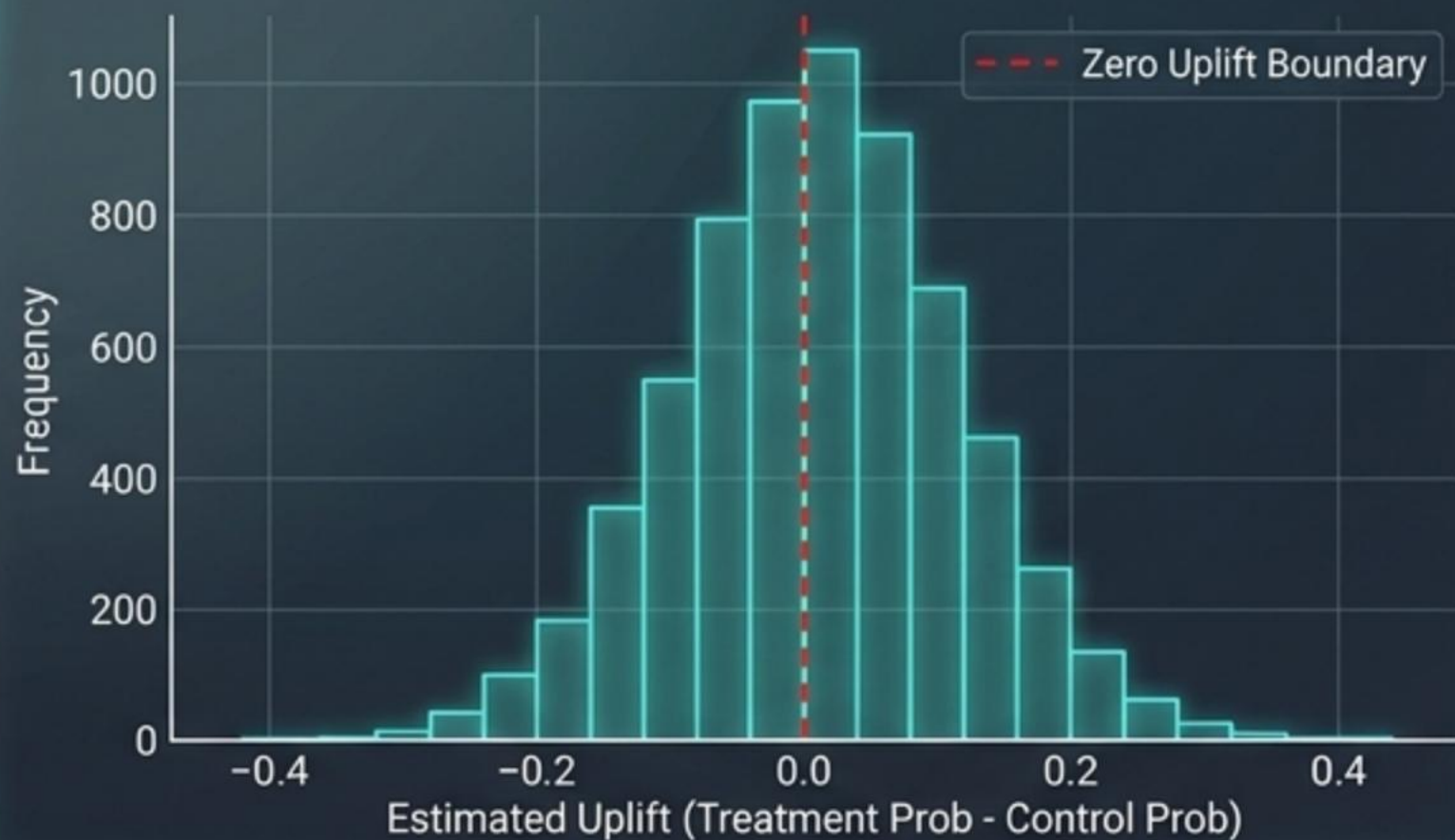
Algorithmic Recourse: Microsoft DiCE



Transitioning from passive classification to algorithmic user agency by mapping minimal counterfactual boundaries.

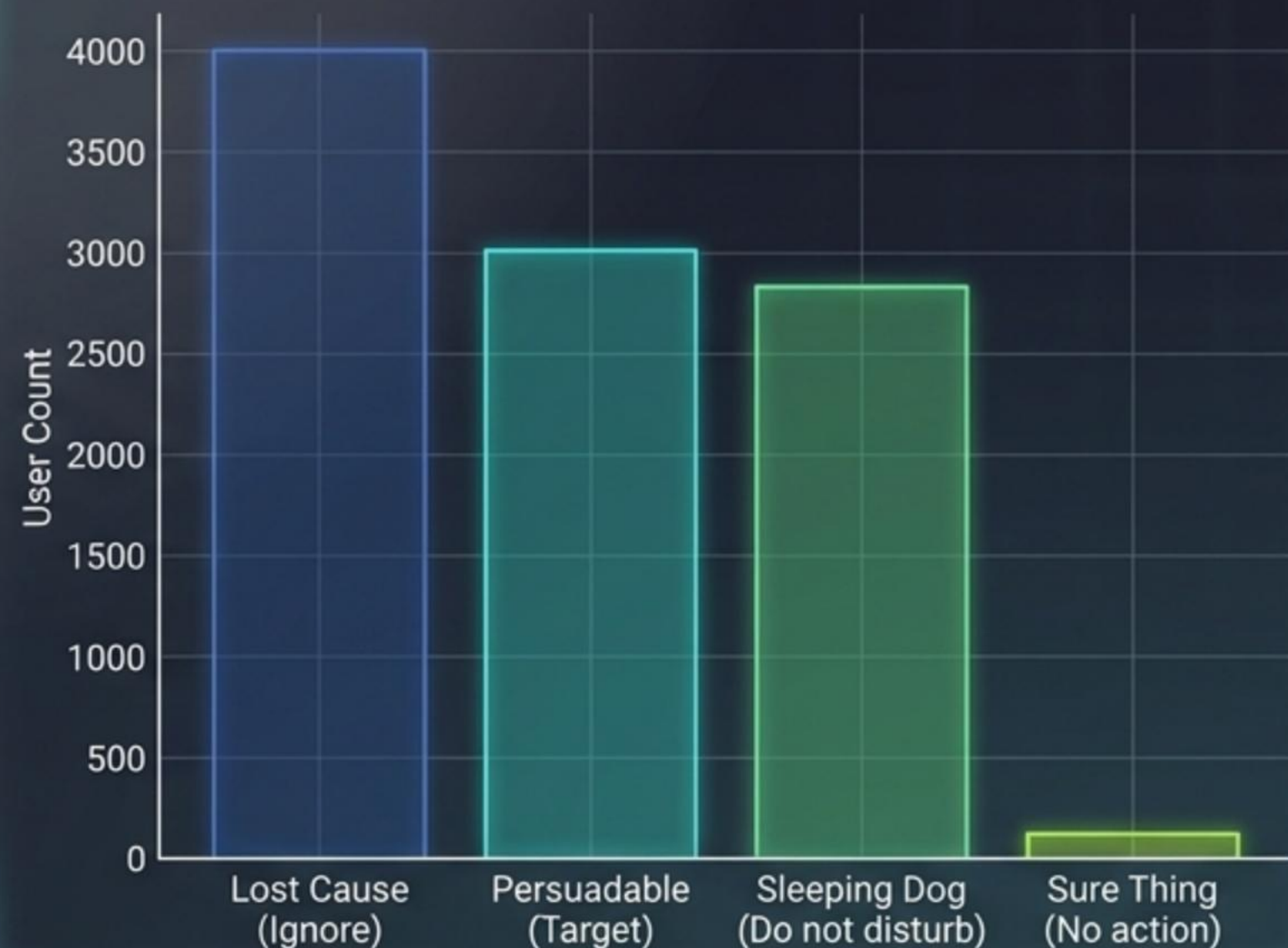
Causal Uplift Segment Targeting

Causal Uplift Score Distribution (ITE)



Average Treatment Effect: 0.0104. The mathematical reality is that static profile metrics carry no significant causal effect.

App Targeting Segments (Causal Prescriptive)



Engine: T-Learner Meta-Classifler / Bias Elimination: K-Fold Cross-Fitting.

Empirical Proof of the Absence of Predictive Signal

ROC-AUC ≈ 0.50

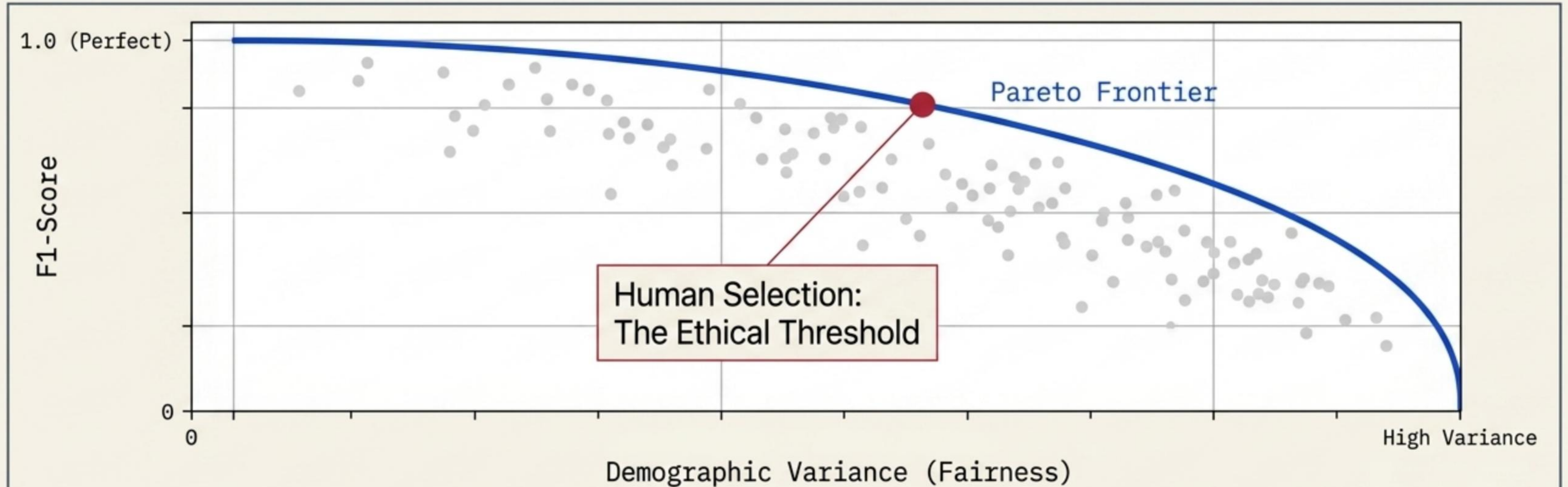
60.30%
Baseline



The Ultimate Convergence: All 16 models—from linear baselines to zero-shot tabular transformers—converge identically at ~60.30% test accuracy. The mathematical equivalent of a coin flip.

Scientific Honesty over Overfitting: There is absolutely no predictive signal in superficial profile features. Love cannot be predicted by swipe ratios, body types, or zodiac signs.

The Ethical Calculus of Predictive Power Versus Fairness



Conflicting Objectives

Maximizing predictive power (F1) inherently risks escalating algorithmic demographic bias.

The Pareto Frontier

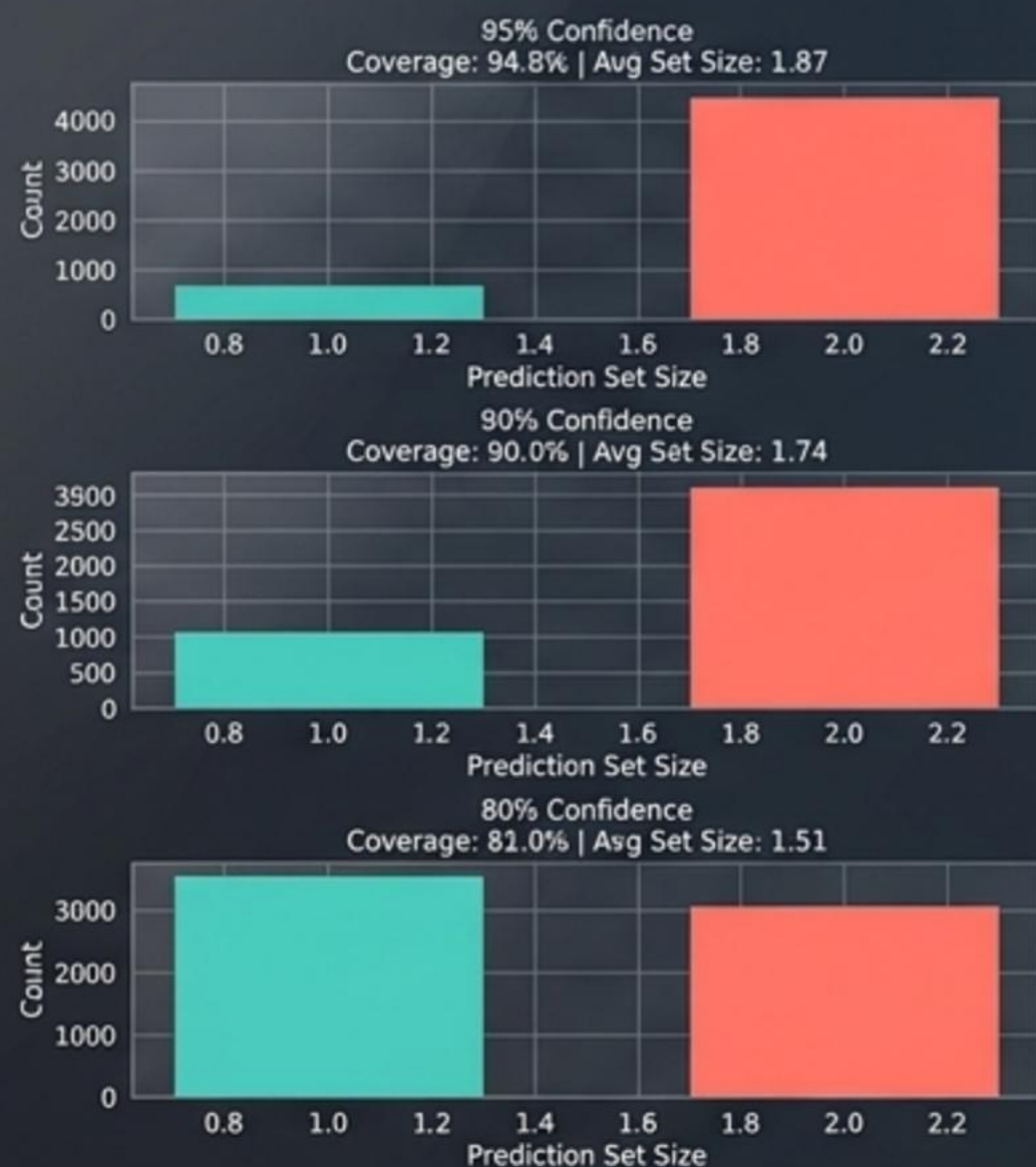
Mapping the mathematical boundary where neither metric can improve without inherently degrading the other.

Ethical AI Engineering

Explicitly selecting an operating threshold that guarantees strict demographic parity across all user sub-groups.

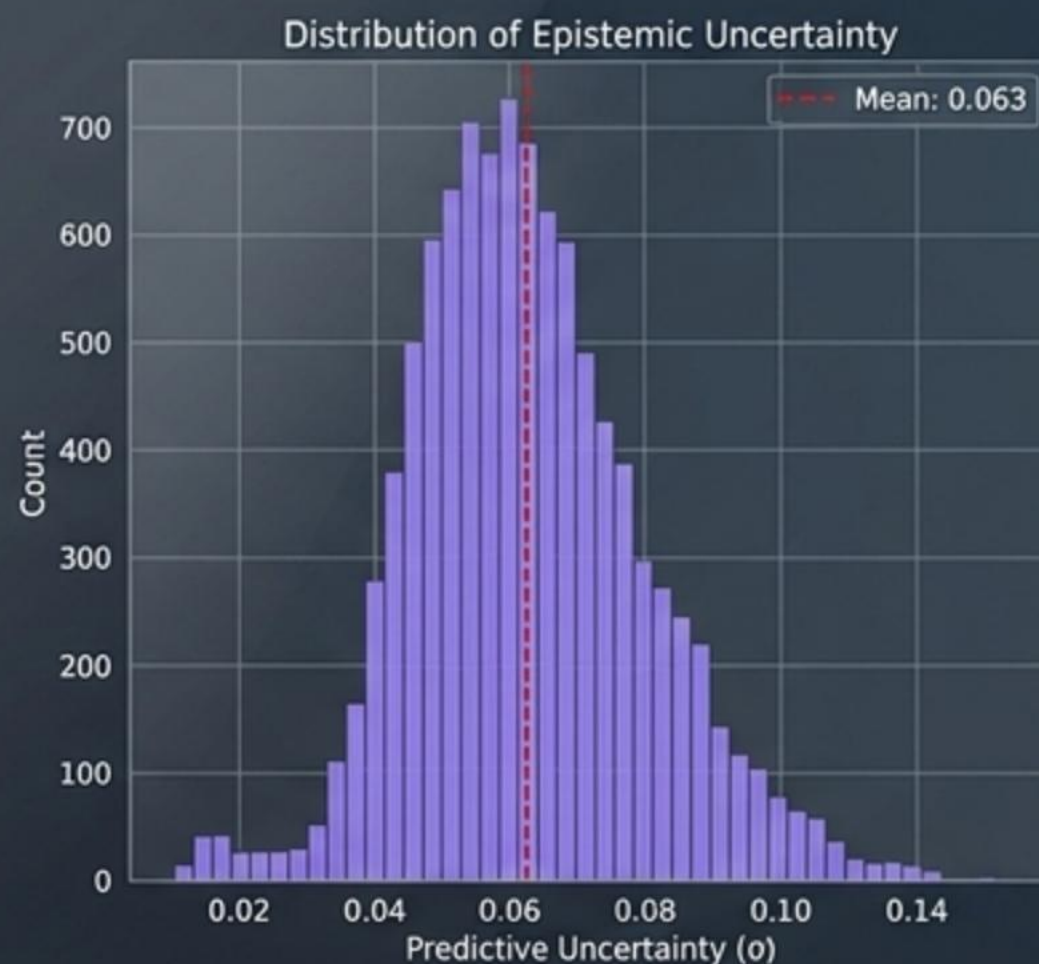
Trustworthy AI: Uncertainty & Fairness

Mathematical Coverage



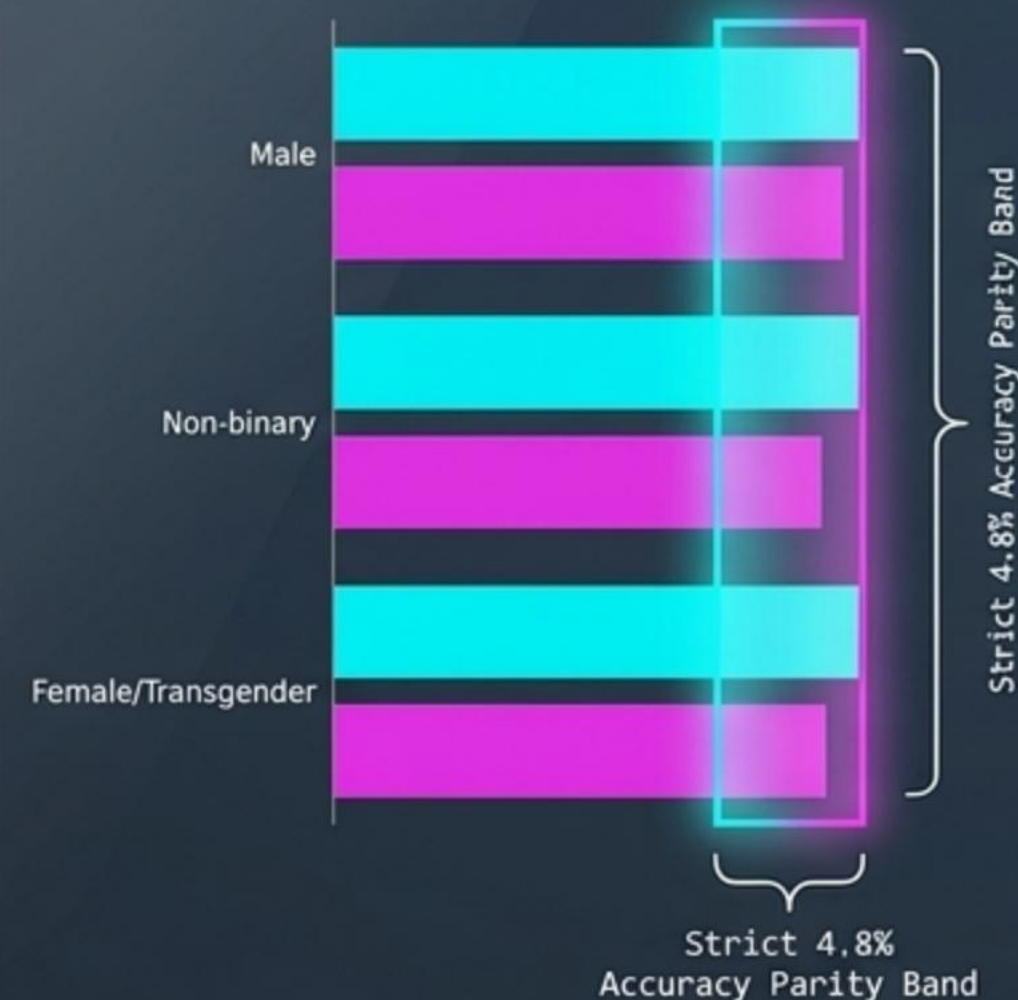
To maintain a 95% theoretical coverage guarantee, the average prediction set size expands to 1.87.

Epistemic Uncertainty



Monte Carlo Dropout reveals predictive confidence is uniformly uncertain near the decision boundary.

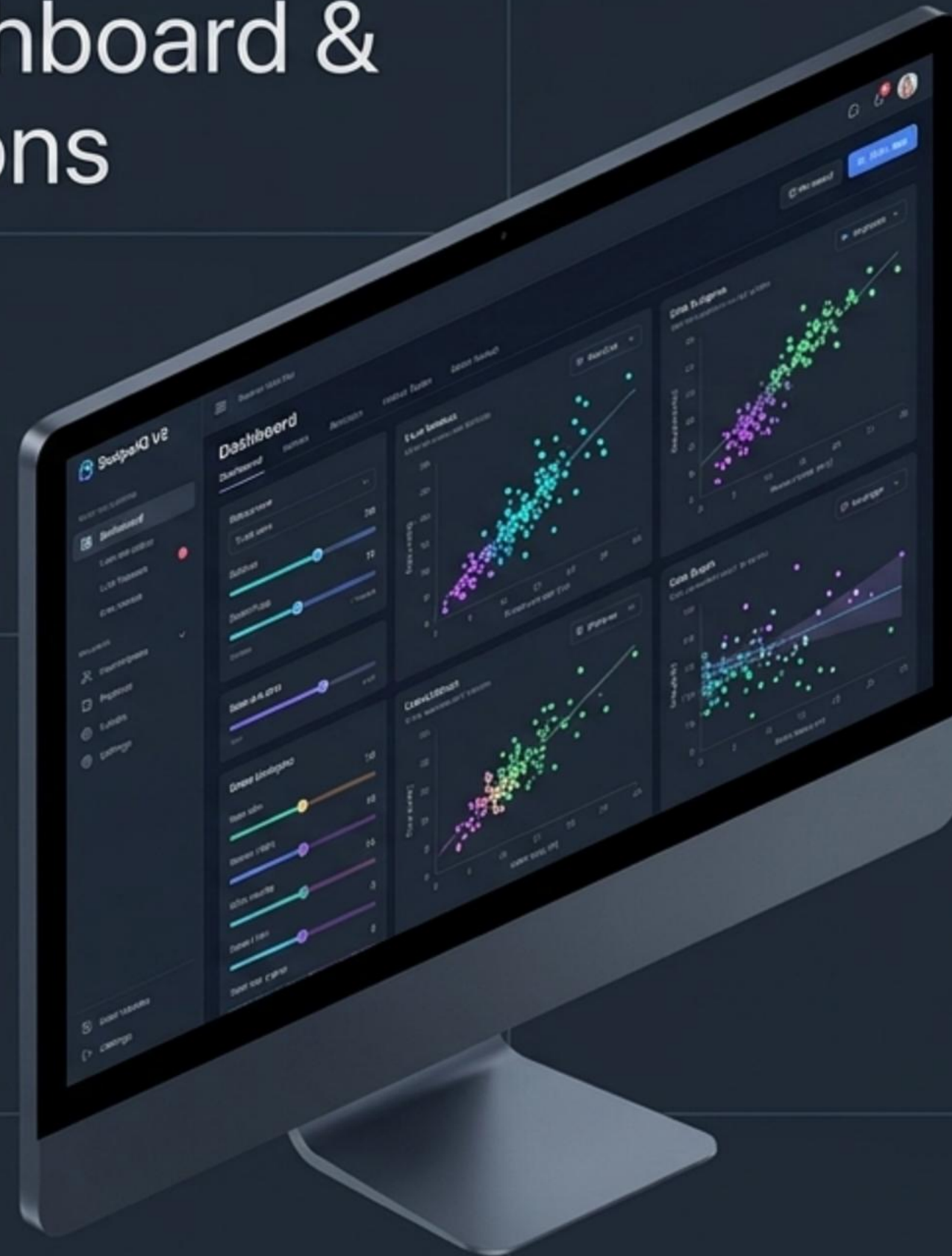
Fairness Audit



Mathematical guarantee of demographic neutrality.

SwipeIQ V2 Dashboard & Recommendations

- 9 Interactive Playgrounds
- Dual-Caching Architecture (<200ms rendering)
- Live Cloud Enterprise Deployment



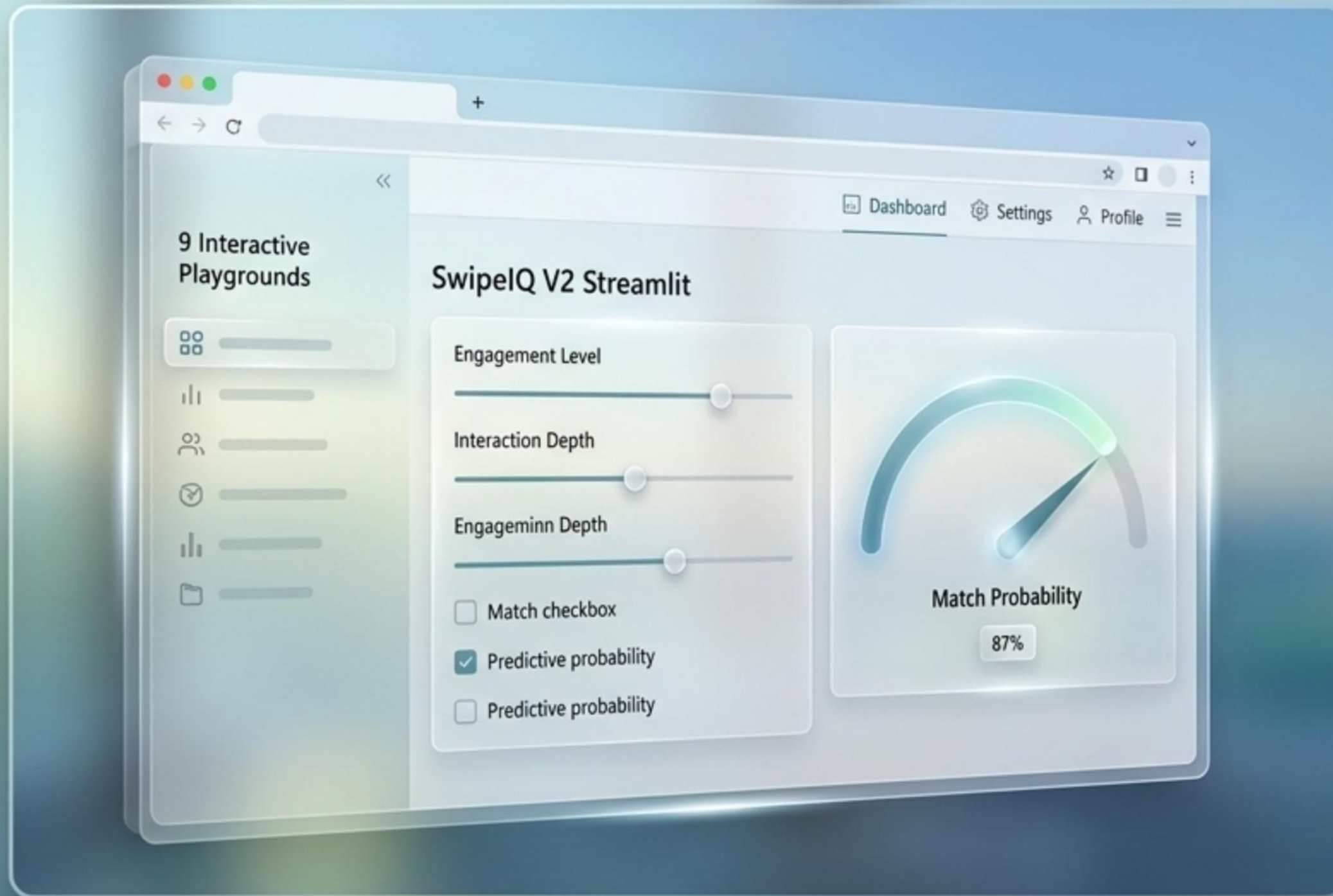
Pivot Matrix

~~Static Metrics
(Zodiac, Demographics,
App Usage)~~

Dynamic Signals
(NLP Bios, Behavioral
Latency)

A complete **paradigm shift**
away from superficial data.

SwipeIQ V2 Dashboard & Recommendations



~~Abandon Static Profile Metrics~~

Zodiac Signs, Swipe Ratios, Static Age

Shift to Behavioral & NLP Signals

Natural Language Bios, Response Latency, Emoji Usage Rates



Live Enterprise Auditing Dashboard: <https://ml-tying-the-data-knot-swipeiq-app.streamlit.app/>

THANK YOU.

The Science of Connection: A Final Note



**ROC-AUC $\approx$ 0.50:
The Mathematical Limit.**

Performance no better
than random guessing.

**“Love cannot
be scientifically
predicted.”**



**Zero Signal in
Superficial Features**

Zodiac signs, swipe ratios,
static metrics carry no
genuine correlation.



**Authentic human
connection remains a
'black box' that eludes
simple tabular data.**